

Surface Integrals — Example 4

~~It's said~~ take note of questions when boundary is rect. or Δ . $\Rightarrow$ for FINALS!

Example

- Let S denote the top hemisphere with center $(0, 0, 0)$ and radius 6.
- Let S be oriented upwards.
- Define a continuous space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (0, 0, z)$.

Problem: Find the flux of $\mathbf{F}$ across S .

Solution

$$\mathcal{S} = \{ (x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 36 \text{ and } z = \sqrt{36 - (x^2 + y^2)} \}$$

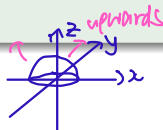
Define a vector function $\vec{r}: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$\vec{r}(\theta, \phi) = \langle 6 \cos \theta \sin \phi, 6 \sin \theta \sin \phi, 6 \cos \phi \rangle \quad \rho = 6 \text{ in this case}$$

For all $\theta, \phi \in \mathbb{R}$, we have

$$\frac{\partial \vec{r}}{\partial \theta}(\theta, \phi) = \langle -6 \sin \theta \sin \phi, 6 \cos \theta \sin \phi, 0 \rangle,$$

$$\frac{\partial \vec{r}}{\partial \phi}(\theta, \phi) = \langle 6 \cos \theta \cos \phi, 6 \sin \theta \cos \phi, -6 \sin \phi \rangle$$



Surface Integrals — Example 4 (cont'd)

$$\left(\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} \right)(\theta, \phi) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -b \sin \theta \sin \phi & b \cos \theta \sin \phi & 0 \\ b \cos \theta \cos \phi & b \sin \theta \cos \phi & -b \sin \theta \end{vmatrix}$$

to check if upwards, look at z component
always +ve for $0 \leq \phi \leq \frac{\pi}{2}$

$$= \langle -3b \cos \theta \sin^2 \phi, -3b \sin \theta \sin^2 \phi, -3b \cos \theta \sin \phi \rangle$$

-ve, so wrong orientation
 $\Rightarrow$ need include -ve sign

$(\vec{r}, [0, 2\pi] \times [0, \frac{\pi}{2}])$ is an orientation-reversing surface parametrization of S .

Flux across S is

$$\iint_S \vec{F} \cdot d\vec{S} = - \iint_{[0, 2\pi] \times [0, \frac{\pi}{2}]} \vec{F}(b \cos \theta \sin \phi, b \sin \theta \sin \phi, b \cos \theta) \cdot \langle -3b \cos \theta \sin^2 \phi, -3b \sin \theta \sin^2 \phi, -3b \cos \theta \sin \phi \rangle dA(\theta, \phi)$$

don't need care as it's zero

$$= - \iint_{[0, 2\pi] \times [0, \frac{\pi}{2}]} (-21b) \cos^2 \phi \sin \phi dA(\theta, \phi) \Rightarrow \text{simplified as } (0, 0, z)$$

$$= 21b \iint_{[0, 2\pi] \times [0, \frac{\pi}{2}]} (\cos^2 \phi \sin \phi) dA(\theta, \phi) \Rightarrow \text{separable}$$

$$= 21b \left(\int_0^{2\pi} 1 d\theta \right) \left(\int_0^{\frac{\pi}{2}} \cos^2 \phi \sin \phi d\phi \right)$$

$$= 21b (2\pi) \left[-\frac{1}{3} \cos^3 \phi \right]_0^{\frac{\pi}{2}}$$

$$= 21b (2\pi) \left(\frac{1}{3} \right)$$

$$= 144\pi$$

Example

- Let S denote the surface boundary of the closed region in $\mathbb{R}^3$ that is
 - bounded above by the top hemisphere given by the equation $z = \sqrt{8 - x^2 - y^2}$;
 - bounded below by the closed cone given by the equation $z = \sqrt{x^2 + y^2}$.
- Let S be oriented positively. $\rightarrow$ outward, away from object
- Define a continuous space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (y, 0, 2)$.

Problem: Find the flux of $\mathbf{F}$ across S .

Solution

pointing outward means pointing upwards

constant ρ

Define $\vec{r}_1, \vec{r}_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$\vec{r}_1(\rho, \theta) = \left\langle \frac{1}{\sqrt{2}}\rho \cos \theta, \frac{1}{\sqrt{2}}\rho \sin \theta, \frac{1}{\sqrt{2}}\rho \right\rangle$$

$$\vec{r}_2(\theta, \rho) = \left\langle \sqrt{8} \cos \theta \sin \phi, \sqrt{8} \sin \theta \sin \phi, \sqrt{8} \cos \phi \right\rangle$$

$\rightarrow$ if pointing outward, it's like pointing downward.

$$\frac{\partial \vec{r}_1}{\partial \rho}(\rho, \theta) = \left\langle \frac{1}{\sqrt{2}} \cos \theta, \frac{1}{\sqrt{2}} \sin \theta, \frac{1}{\sqrt{2}} \right\rangle$$

$$\frac{\partial \vec{r}_1}{\partial \theta}(\rho, \theta) = \left\langle -\frac{1}{\sqrt{2}} \rho \sin \theta, \frac{1}{\sqrt{2}} \rho \cos \theta, 0 \right\rangle$$

$\phi = 45^\circ$

Surface Integrals — Example 5 (cont'd)

$$\frac{\partial \vec{r}_2}{\partial \theta}(\theta, \phi) = \langle -\sqrt{8} \sin \theta \sin \phi, \sqrt{8} \cos \theta \sin \phi, 0 \rangle$$

$$\frac{\partial \vec{r}_2}{\partial \phi}(\theta, \phi) = \langle \sqrt{8} \cos \theta \cos \phi, \sqrt{8} \sin \theta \cos \phi, -\sqrt{8} \sin \theta \rangle$$

$$\left(\frac{\partial \vec{r}_1}{\partial \rho} \times \frac{\partial \vec{r}_1}{\partial \theta} \right)(\rho, \theta) = \langle -\frac{1}{2} \rho \cos \theta, -\frac{1}{2} \rho \sin \theta, \frac{1}{2} \rho \rangle$$

$$\left(\frac{\partial \vec{r}_2}{\partial \theta} \times \frac{\partial \vec{r}_2}{\partial \phi} \right)(\theta, \phi) = \langle -8 \cos \theta \sin^2 \phi, -8 \sin \theta \sin^2 \phi, -8 \cos \theta \sin \phi \rangle$$

$(\vec{r}_1, [0, \sqrt{8}] \times [0, 2\pi])$ is an orientation reversing surface parametrization of S_1 .

$(\vec{r}_2, [0, 2\pi] \times [0, \frac{\pi}{4}])$ is an orientation reversing surface parametrization of S_2 .

$$\iint_{S_1} \vec{F} \cdot d\vec{S} = - \iint_{[0, \sqrt{8}] \times [0, 2\pi]} \langle \frac{1}{\sqrt{2}} \rho \sin \theta, 0, 2 \rangle \cdot \langle -\frac{1}{2} \rho \cos \theta, -\frac{1}{2} \rho \sin \theta, \frac{1}{2} \rho \rangle dA(\rho, \theta)$$

$$= - \iint_{[0, \sqrt{8}] \times [0, 2\pi]} \left[-\frac{1}{2\sqrt{2}} \rho^2 \sin \theta \cos \theta + \rho \right] dA(\rho, \theta)$$

$$= - \int_0^{2\pi} \left[\int_0^{\sqrt{8}} \left[-\frac{1}{4\sqrt{2}} \rho^2 \sin 2\theta + \rho \right] d\rho \right] d\theta$$

$$= \dots$$

$$= -8\pi$$

$$\iint_{S_2} \vec{F} \cdot d\vec{S} = - \iint_{[0, 2\pi] \times [0, \frac{\pi}{4}]} \langle \sqrt{8} \sin \theta \sin \phi, 0, 2 \rangle \cdot \langle \dots \rangle dA(\theta, \phi)$$

$$= - \iint_{[0, 2\pi] \times [0, \frac{\pi}{4}]} \left[-8\sqrt{8} \cos \theta \sin \theta \sin^2 \phi - 16 \cos \theta \sin \phi \right] dA(\theta, \phi)$$

$$= \dots 8\pi$$

$$\iint_S \vec{F} \cdot d\vec{S} = (-8\pi) + 8\pi = 0$$

Surface Integrals — Example 6

Example

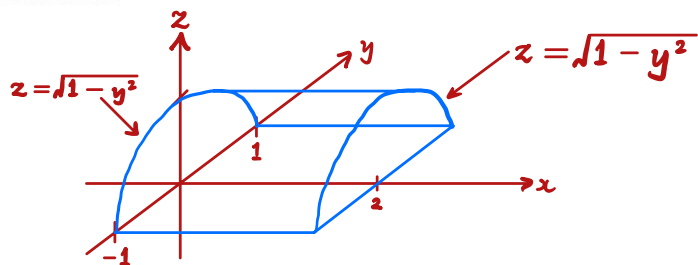
- Let S denote the surface boundary of the closed region E in $\mathbb{R}^3$ specified by

$$E \stackrel{\text{df}}{=} \left\{ (x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq 2, \quad -1 \leq y \leq 1, \quad \text{and } 0 \leq z \leq \sqrt{1 - y^2} \right\}.$$

- Let S be oriented positively.
- Define a continuous space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (x^2, y^2, z^2)$.

Problem: Find the flux of $\mathbf{F}$ across S .

Solution



- Instead of evaluating four separate surface integrals, we will apply the Gauss-Ostrogradsky Divergence Theorem introduced at the end of this lecture.

Surface Integrals — Example 6 (cont'd)

- For all $x, y, z \in \mathbb{R}$, we have

$$[\operatorname{div}(\vec{F})](x, y, z) = \frac{\partial}{\partial x}[x^2] + \frac{\partial}{\partial y}[y^2] + \frac{\partial}{\partial z}[z^2] = 2x + 2y + 2z = 2(x + y + z).$$

- Let D denote the region in the yz -plane that is specified by $\{(y, z) \in \mathbb{R}^2 \mid -1 \leq y \leq 1 \text{ and } 0 \leq z \leq \sqrt{1 - y^2}\}$.
- Then E is a Type-II region in $\mathbb{R}^3$ that is specified by $\{(x, y, z) \in \mathbb{R}^3 \mid (y, z) \in D \text{ and } 0 \leq x \leq 2\}$.

- Therefore, by the Gauss-Ostrogradsky Divergence Theorem,

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \iiint_E \operatorname{div}(\vec{F}) \, dV \\ &= \iiint_E 2(x + y + z) \, dV(x, y, z) \\ &= 2 \iiint_E (x + y + z) \, dV(x, y, z) \\ &= 2 \iint_D \left[\int_0^2 (x + y + z) \, dx \right] dA(y, z) \\ &= 2 \iint_D \left[\frac{1}{2}x^2 + x(y + z) \right]_{x=0}^{x=2} dA(y, z) \\ &= 2 \iint_D 2(1 + y + z) \, dA(y, z) \end{aligned}$$

Surface Integrals — Example 6 (cont'd)

$$\begin{aligned} &= 4 \iint_D (1 + y + z) \, dA(y, z) \\ &= 4 \iint_{[0,1] \times [0,\pi]} [1 + r \cos(\theta) + r \sin(\theta)] \cdot r \, dA(r, \theta) \\ &= 4 \iint_{[0,1] \times [0,\pi]} [r + r^2 \cos(\theta) + r^2 \sin(\theta)] \, dA(r, \theta) \\ &= 4 \int_0^\pi \left[\int_0^1 [r + r^2 \cos(\theta) + r^2 \sin(\theta)] \, dr \right] d\theta \\ &= 4 \int_0^\pi \left[\frac{1}{2} r^2 + \frac{1}{3} r^3 \cos(\theta) + \frac{1}{3} r^3 \sin(\theta) \right]_{r=0}^{r=1} d\theta \\ &= 4 \int_0^\pi \left[\frac{1}{2} + \frac{1}{3} \cos(\theta) + \frac{1}{3} \sin(\theta) \right] d\theta \\ &= 4 \left[\frac{1}{2} \theta + \frac{1}{3} \sin(\theta) - \frac{1}{3} \cos(\theta) \right]_0^\pi \\ &= 4 \left[\left(\frac{\pi}{2} + \frac{1}{3} \right) - \left(-\frac{1}{3} \right) \right] \\ &= 4 \left(\frac{\pi}{2} + \frac{2}{3} \right) \\ &= 2\pi + \frac{8}{3} \cdot \otimes \end{aligned}$$

We will cover only the following four topics in the Week 12 lectures:

- 1 The curl of a space vector field (only for dimension 3).
- 2 The divergence of a vector field (of any dimension).
- 3 Stokes' Theorem. *Tested*
- 4 The Gauss-Ostrogradsky Divergence Theorem (not tested).

Vector Fields — Curl

One of the most important operations on space vector fields is the **curl**.

Dimension 3

Definition (Curl of a Space Vector Field)

Let $\mathbf{F} = (P, Q, R)$ be a space vector field.

3 Components

The **curl** of $\mathbf{F}$, denoted by $\text{curl}(\mathbf{F})$, is the space vector field defined as follows:

$$\text{curl}(\mathbf{F}) \stackrel{\text{df}}{=} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right).$$

→ 3x3 determinant

Remark: $\text{Dom}(\text{curl}(\mathbf{F})) \subseteq \text{Dom}(\mathbf{F})$. In most cases, these two domains are equal.

The curl of a space vector field is a space vector field.

Vector Fields — Curl

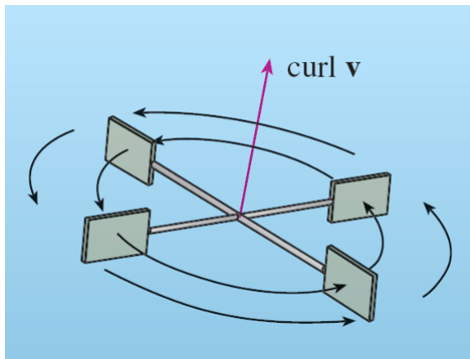
The curl of a space vector field has a physical interpretation.

Let $\mathbf{v}$ represent the velocity field of a fluid.

Place a paddlewheel at a point P in the fluid, with its axis aligned with $[\text{curl}(\mathbf{v})](P)$.

rotational power

Then $[\text{curl}(\mathbf{v})](P)$ measures the propensity of the paddlewheel to spin counterclockwise under the influence of the fluid.



The curl was already introduced in Week 9 as an ingredient of the following test.

Theorem (Test for Conservativeness of a Space Vector Field)

Let $\mathbf{F}$ be a space vector field whose domain is open and simply connected. } e.g. $\mathbb{R}^3$, open unit ball
no holes

If $\mathbf{F}$ is smooth, then

$$\mathbf{F} \text{ is conservative} \iff \operatorname{curl}(\mathbf{F}) \equiv (0, 0, 0) \text{ on } \operatorname{Dom}(\mathbf{F}).$$

This works only for three dimensions

Example

Define a space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (x^2z, e^y + xz, xyz)$.

Problem: Compute the curl of $\mathbf{F}$.

Solution

$$\begin{aligned} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2z & e^y + xz & xyz \end{vmatrix} &= \langle \frac{\partial}{\partial y}[xyz] - \frac{\partial}{\partial z}[e^y + xz], \frac{\partial}{\partial z}[x^2z] - \frac{\partial}{\partial x}[xyz], \frac{\partial}{\partial x}[e^y + xz] - \frac{\partial}{\partial y}[x^2z] \rangle \\ &= \langle xz - x, x^2 - yz, z \rangle \end{aligned}$$

For all $x, y, z \in \mathbb{R}$, we have $[\text{curl}(\mathbf{F})](x, y, z) = \langle xz - x, x^2 - yz, z \rangle$

The second most important operation on vector fields is the **divergence**.

Definition (Divergence of an n -Dimensional Vector Field)

Let $n \in \mathbb{N}$, and let $\mathbf{F} = (F_1, \dots, F_n)$ be an n -dimensional vector field.

$\hookrightarrow n$ components of $\vec{F}$

The **divergence** of $\mathbf{F}$, denoted by $\text{div}(\mathbf{F})$, is the scalar field defined as follows:

$$\text{div}(\mathbf{F}) \stackrel{\text{df}}{=} \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right) \bullet (F_1, \dots, F_n) = \frac{\partial F_1}{\partial x_1} + \dots + \frac{\partial F_n}{\partial x_n}.$$

The n -tuple $\left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)$ is usually denoted by “ ∇ ” and called the “**del operator**”; for this reason, it is very common to see $\text{div}(\mathbf{F})$ written as “ $\nabla \bullet \mathbf{F}$ ”.

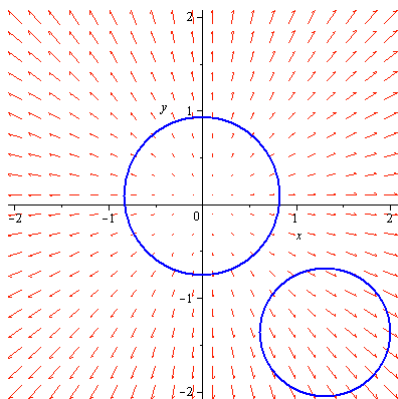
$\hookrightarrow$ gradient

Remarks:

- $\text{Dom}(\text{div}(\mathbf{F})) \subseteq \text{Dom}(\mathbf{F})$. *$\hookleftarrow$ In most cases, these 2 domains are equal.*
- When we apply ∇ to a scalar field f , we get the gradient of f .

In three dimensions: Given $\vec{F} = (P, Q, R)$, we have

$$\text{div}(\vec{F}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

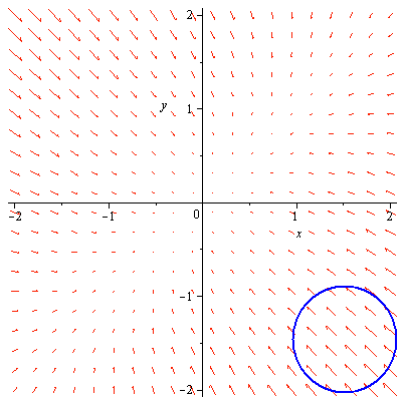


The plane vector field $\mathbf{F}$ depicted here is defined on $\mathbb{R}^2$ by $\mathbf{F}(x, y) \stackrel{\text{df}}{=} (x, y)$.
 $[\text{div}(\mathbf{F})](x, y) = \frac{\partial}{\partial x} [x] + \frac{\partial}{\partial y} [y] = 1 + 1 = 2$

Note that $\text{div}(\mathbf{F}) \equiv 2 > 0$ on $\mathbb{R}^2$.

This means that there is a net outflow through every circle.

Vector Fields — Divergence

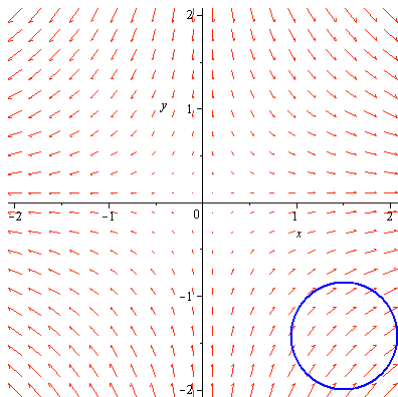


The plane vector field $\mathbf{F}$ depicted here is defined on $\mathbb{R}^2$ by $\mathbf{F}(x, y) \stackrel{\text{df}}{=} (y - 2x, x - 2y)$.

Note that $\text{div}(\mathbf{F}) \equiv -4 < 0$ on $\mathbb{R}^2$.

This means that there is a net inflow through every circle.

Vector Fields — Divergence



The plane vector field $\mathbf{F}$ depicted here is defined on $\mathbb{R}^2$ by $\mathbf{F}(x, y) \stackrel{\text{df}}{=} (x, -y)$.

Note that $\text{div}(\mathbf{F}) \equiv 0$ on $\mathbb{R}^2$.

there can still be movement!
→ they just cancel out so net zero.

This means that there is zero net flow through every circle.

The divergence of a vector field $\mathbf{F}$ at a point P can be interpreted physically as follows:

- $[\operatorname{div}(\mathbf{F})](P)$ is positive: There is a net outflow locally around P .
- $[\operatorname{div}(\mathbf{F})](P)$ is negative: There is a net inflow locally around P .
- $[\operatorname{div}(\mathbf{F})](P)$ is zero: There is zero net flow locally around P .

Example

Define a space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (e^x, yz, -yz^2)$.

Problem: Evaluate the divergence of $\mathbf{F}$ at the point $(0, 2, -1)$.

$$\frac{\partial}{\partial x} [e^x] + \frac{\partial}{\partial y} [yz] + \frac{\partial}{\partial z} [-yz^2] = e^x + z - 2yz$$

For all $x, y, z \in \mathbb{R}$, we have $[\text{div}(\mathbf{F})](x, y, z) = e^x + z - 2yz$

$$[\text{div}(\mathbf{F})](0, 2, -1) = e^0 + (-1) - 2(2)(-1) = 1 - 1 + 4 = 4$$

Vector Fields — Gradient, Curl, and Divergence

Let U be an open subset of $\mathbb{R}^3$, and let $\mathbf{F}$ be a space vector field on U .

Suppose that the three component functions of $\mathbf{F}$ have continuous second-order partial derivatives that are defined throughout U .

scalar field
vector field

Then you have to show, in Week 13 Tutorial Problem Set, that $\text{div}(\text{curl}(\mathbf{F})) \equiv 0$ on U .

Definition (Divergence-Free Space Vector Fields)

A vector field is said to be **divergence-free** if and only if its divergence is the zero scalar function on its domain.

Hence, $\text{curl}(\mathbf{F})$ is divergence-free.

Partial derivatives of all orders exist

Let $\text{ScalarFields}(U)$ denote the $\mathbb{R}$ -vector space of C^∞ space scalar fields on U .

Let $\text{VectorFields}(U)$ denote the $\mathbb{R}$ -vector space of C^∞ space vector fields on U .

Then we have the following sequence of linear maps, known as a co-chain complex:

not tested

$$\text{ScalarFields}(U) \xrightarrow{\text{grad}} \text{VectorFields}(U) \xrightarrow{\text{curl}} \text{VectorFields}(U) \xrightarrow{\text{div}} \text{ScalarFields}(U).$$

Linear Transformations

Stokes' Theorem

Stokes' Theorem is a powerful tool that allows us to do the following:

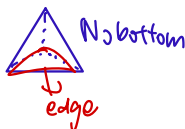
- Compute a complicated loop integral in 3-D space as an easier surface integral.
- Compute a complicated surface integral as an easier loop integral in 3-D space.

Theorem (Stokes' Theorem)

- Let S be an oriented piecewise-smooth surface made up of finitely many oriented, nice, and smooth parametric surfaces intersecting only along their edges such that the net edge of S is a simple loop C in $\mathbb{R}^3$ whose length exists.
- Suppose that C is positively oriented with respect to the orientation of S .
- Let $\mathbf{F}$ be a smooth space vector field such that $S \subseteq \text{Dom}(\mathbf{F})$.

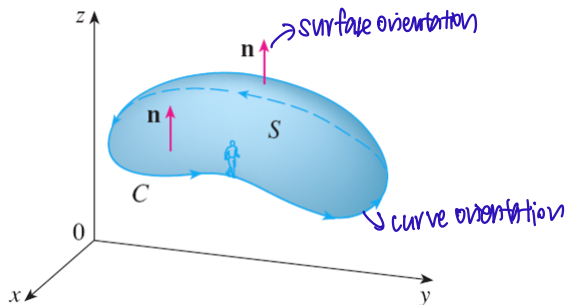
Then $\iint_S \text{curl}(\mathbf{F}) \bullet d\mathbf{S} = \oint_C \mathbf{F} \bullet d\mathbf{r}$. ~~XXXX~~

surface integral of a vector field loop integral of a vector field



Stokes' Theorem — Orientation

Before we do examples, a few things must be said about orientation.



Let $\mathbf{n}$ be a given orientation of S .

The positive orientation of the curve C with respect to $\mathbf{n}$ is designated as the direction of travel along C that puts S on your left while your head points in the direction of $\mathbf{n}$.

Alternatively, the positive orientation of C is provided by the fingers of the Right-Hand Grip Rule, with the thumb pointing in the direction of $\mathbf{n}$.

Stokes' Theorem — Example 1

Example

- Let C denote the oriented curve in $\mathbb{R}^3$ that is the intersection of
 - the cylinder given by the equation $x^2 + y^2 = 1$ and
 - the plane given by the equation $y + z = 2$,where the orientation is counterclockwise when viewed from above.
- Define a smooth space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (-y^2, x, z^2)$.

Problem: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Line Integral

upward $\rightarrow$ z component positive

- Let π denote the plane given by the equation $y + z = 2$
- Let S denote the part of π that is enclosed by C .

find a surface parametrization of S .

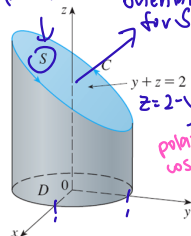
- Define a vector function $\vec{r}: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$\vec{r}(r, \theta) = \langle r \cos \theta, r \sin \theta, 2 - r \sin \theta \rangle$$

$$\frac{\partial \vec{r}}{\partial r}(r, \theta) = \langle \cos \theta, \sin \theta, -\sin \theta \rangle$$

$$\frac{\partial \vec{r}}{\partial \theta}(r, \theta) = \langle -r \sin \theta, r \cos \theta, -r \cos \theta \rangle$$

Solution
part of π



orientation
for S

use
polar coord.
cos circle!

Stokes' Theorem — Example 1 (cont'd)

$$\left(\frac{\partial \vec{r}}{\partial r} \times \frac{\partial \vec{r}}{\partial \theta} \right) (r, \theta) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \theta & \sin \theta & -\sin \theta \\ -r \sin \theta & r \cos \theta & -r \cos \theta \end{vmatrix} = \langle 0, r, r \rangle \quad \text{orientation is correct.}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & x & z^2 \end{vmatrix} = \langle 0, 0, 1+2y \rangle$$

For all $x, y, z \in \mathbb{R}$, we have $[\text{curl}(\vec{F})](x, y, z) = \langle 0, 0, 1+2y \rangle$

$(\vec{r}, [0, 1] \times [0, 2\pi])$ is a surface parametrization of S .

By Stokes' Theorem, $\downarrow$ put $\vec{r}(r, \theta)$ into $\text{curl}(\vec{F})$, NOT $\vec{F}$ itself

$$\begin{aligned} \int_S \vec{F} \cdot d\vec{r} &= \iint_S [\text{curl}(\vec{F})] \cdot d\vec{r} \\ &= \iint_{[0, 1] \times [0, 2\pi]} [\text{curl}(\vec{F})](\overset{x}{r \cos \theta}, \overset{y}{r \sin \theta}, \overset{z}{2 - r \sin \theta}) \cdot \langle 0, r, r \rangle \, dA(r, \theta) \\ &= \iint_{[0, 1] \times [0, 2\pi]} \langle 0, 0, 1+2r \sin \theta \rangle \cdot \langle 0, r, r \rangle \, dA(r, \theta) \\ &= \iint_{[0, 1] \times [0, 2\pi]} [r + 2r^2 \sin \theta] \, dA(r, \theta) \quad \text{not separable} \end{aligned}$$

Stokes' Theorem — Example 1 (cont'd)

$$\begin{aligned} &= \int_0^{2\pi} \left[\int_0^1 (r + 2r^2 \sin \theta) dr \right] d\theta \\ &= \int_0^{2\pi} \left[\frac{r^2}{2} + \frac{2r^3}{3} \sin \theta \right]_{r=0}^{r=1} d\theta \\ &= \int_0^{2\pi} \left[\frac{1}{2} + \frac{2}{3} \sin \theta \right] d\theta \\ &= \left[\frac{1}{2} \theta - \frac{2}{3} \cos \theta \right]_0^{2\pi} \\ &= \pi \end{aligned}$$

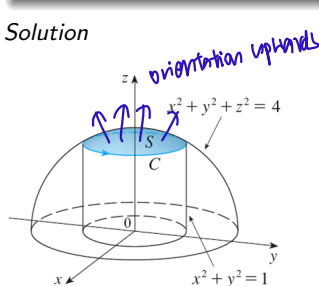
Stokes' Theorem — Example 2

Example

- Let S denote the part of the sphere with center $(0, 0, 0)$ and radius 2 that lies
 - inside the cylinder given by the equation $x^2 + y^2 = 1$ and
 - above the xy -plane,oriented upwards.
- Define a smooth space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by $\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (xz, yz, xy)$.

Problem: Use Stokes' Theorem to evaluate $\iint_S \text{curl}(\mathbf{F}) \bullet d\mathbf{S}$.

Solution



For all $x, y, z \in \mathbb{R}$, we have

$$\begin{cases} x^2 + y^2 = 1 \\ x^2 + y^2 + z^2 = 4 \\ z > 0 \end{cases} \iff \begin{cases} x^2 + y^2 = 1 \\ z = \sqrt{3} \end{cases}$$

Define a vector function $\vec{r}: [0, 2\pi] \rightarrow \mathbb{R}^3$ by

$$\vec{r}(t) = \langle \cos t, \sin t, \sqrt{3} \rangle$$

C is counter clock wise, correct orientation

- Then $\vec{r}$ is a nice orientation-respecting parametrization of C .
- The orientation of C is positive w.r.t the orientation of S (by the Right Hand Grip Rule)

right hand grip rule satisfied

Stokes' Theorem — Example 2 (cont'd)

By Stokes Theorem,

$$\begin{aligned}\iint_S \text{curl}(\vec{F}) \cdot d\vec{s} &= \oint_C \vec{F} \cdot d\vec{r} \\&= \int_0^{2\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \\&= \int_0^{2\pi} \vec{F}(\cos t, \sin t, \sqrt{3}) \cdot \langle -\sin t, \cos t, 0 \rangle dt \\&= \int_0^{2\pi} \langle \sqrt{3}\cos t, \sqrt{3}\sin t, \cos t \sin t \rangle \cdot \langle -\sin t, \cos t, 0 \rangle dt \\&= \int_0^{2\pi} 0 dt \\&= 0\end{aligned}$$

The Gauss-Ostrogradsky Divergence Theorem

Not tested.

Theorem (The Gauss-Ostrogradsky Divergence Theorem)

- Let E be a bounded and closed region in $\mathbb{R}^3$ with the following two properties:
 - The volume of E exists.
 - The boundary of E is a piecewise-smooth surface S with no edge.
- Let $\mathbf{F}$ be a smooth space vector field such that $E \subseteq \text{Dom}(\mathbf{F})$.
- Suppose that S is oriented positively.

Then
$$\iint_S \mathbf{F} \bullet d\mathbf{S} = \iiint_E \text{div}(\mathbf{F}) \, dV.$$

The Gauss-Ostrogradsky Divergence Theorem allows complicated surface integrals over edgeless piecewise-smooth surfaces to be evaluated more expediently as triple integrals.

The Gauss-Ostrogradsky Divergence Theorem — Example 1

Example

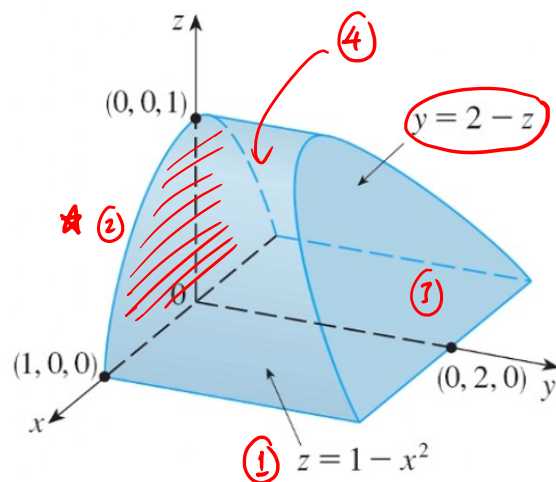
- Let S denote the surface boundary of the closed region in $\mathbb{R}^3$ that is bounded by
 - the parabolic cylinder given by the equation $z = 1 - x^2$ and
 - the three planes given by the equations $y = 0$, $z = 0$, and $y + z = 2$.
- Let S be oriented positively. *Outward.*
- Define a smooth space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} \left(\overset{\textcircled{1}}{xy}, \overset{\textcircled{2}}{y^2 + e^{xz^2}}, \overset{\textcircled{3}}{\sin(xy)} \right).$$

Problem: Use the Gauss-Ostrogradsky Divergence Theorem to evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$.

Solution

The Gauss-Ostrogradsky Divergence Theorem — Example 1 (cont'd)



- Let E denote the closed region in $\mathbb{R}^3$ whose surface boundary is in S .
- Then E can be specified as a Type-III region in $\mathbb{R}^3$ as follows:
 $E = \{(x, y, z) \in \mathbb{R}^3 \mid -1 \leq x \leq 1, 0 \leq z \leq 1 - x^2, 0 \leq y \leq 2 - z\}$.

$$\cdot \frac{\partial}{\partial x}[xy] + \frac{\partial}{\partial y}[y^2 + e^{xz^2}] + \frac{\partial}{\partial z}[\sin(xy)] = y + 2y + 0 = 3y.$$

- Therefore, by the Divergence Theorem,

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \iiint_E \operatorname{div}(\vec{F}) \, dV \\ &= \iiint_E 3y \, dV(x, y, z) \\ &= 3 \iiint_E y \, dV(x, y, z) \\ &= 3 \int_{-1}^1 \left[\int_0^{1-x^2} \left[\int_0^{2-z} y \, dy \right] dz \right] dx \\ &= 3 \int_{-1}^1 \left[\int_0^{1-x^2} \left[\frac{y^2}{2} \right]_{y=0}^{y=2-z} dz \right] dx \end{aligned}$$

The Gauss-Ostrogradsky Divergence Theorem — Example 1 (cont'd)

$$\begin{aligned}
 &= 3 \int_{-1}^1 \left[\int_0^{1-x^2} \frac{1}{2} (2-z)^2 dz \right] dx \\
 &= \frac{3}{2} \int_{-1}^1 \left[\int_0^{1-x^2} (2-z)^2 dz \right] dx \\
 &= \frac{3}{2} \int_{-1}^1 \left[-\frac{(2-z)^3}{3} \right]_{z=0}^{z=1-x^2} dx \\
 &= -\frac{1}{2} \int_{-1}^1 [(2-z)^3]_{z=0}^{z=1-x^2} dx \\
 &= -\frac{1}{2} \int_{-1}^1 [(1+x^2)^3 - 8] dx \\
 &= -\frac{1}{2} \int_{-1}^1 [x^6 + 3x^4 + 3x^2 - 7] dx \\
 &= -\frac{1}{2} \left[\frac{x^7}{7} + \frac{3x^5}{5} + x^3 - 7x \right]_{-1}^1 \\
 &= \frac{184}{35}.
 \end{aligned}$$